COM3031-Probabilistic Machine Learning
Workshop: Q-Learning and Multi-Armed Bandits

Overview

This workshop mainly focuses on two reinforcement learning problems: Q-learning and Multi-Armed Bandits (MAB). You will use the provided Jupyter notebook to understand the implementations and complete the required tasks.

1 Q-Learning

The problem description is given as follows: assume that an agent is placed in an environment consisting of several connected locations. The goal is to use Q-learning to learn the optimal path from a chosen starting location to a target location. The tasks are given as follows,

Instructions

1. Study the provided Jupyter notebook and understand how the Q-learning model is implemented.
2. Identify the locations (states) and the reward matrix used in the environment.
3. Define a new starting location and a new ending location.
4. Use the learned Q-values to find the optimal route from the chosen starting location to the chosen ending location.
5. Report clearly:
    - the chosen start location,
    - the chosen end location,
    - the optimal route returned by the algorithm.

2 Multi-Armed Bandits

The problem description is given as follows: assume that an agent must repeatedly choose between several slot machines (arms), each with an unknown reward distribution. The goal is to maximise the total reward by balancing exploration and exploitation. The tasks are given as follows,

Instructions

1. Study the implementation of the Multi-Armed Bandit problem in the provided notebook.
2. Understand how the ε-greedy strategy is used to balance exploration and exploitation.
3. Run the algorithm and observe how the estimated action values are updated over time.
4. Identify which arm is estimated to be the best after learning.
5. Briefly comment on the effect of the choice of ε on the learning behaviour.